

Route-F Theory Audit: Corrected Flavor Geometry, Vacuum Matching, and Constructive Repair Mechanisms

P54PQ-v2 four-dimensional mainline

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Abstract

This audit corrects a scalar-conjugation error and an overstatement of a tree-level flavor exclusion. The declared action fixes $|r| = 1.10495$, $|s| = 0.0336073$, not the earlier pair $(0.905014, 0.0410320)$. The exact two-matrix identity still gives strong diagnostic tension, but the complete loop-corrected light eigenvector and Pati–Salam Yukawa matching have not been computed. The full model is not thereby excluded.

A missing same-action calculation gives a positive heavy-relaxed quartic $\lambda_{\text{EFT}} = 0.409154$. We derive the Schur-complement light-state condition, the complex Majorana Coleman–Weinberg projection, an antisymmetric-only flavor obstruction, a normalized-overlap feasibility theorem, and a PQ-preserving mirror-10 operator with a vectorlike mediator. The last construction exactly matches a third-family toy, not a global fit. Minimal Einstein–Cartan torsion and central quotients do not automatically supply the missing symmetric flavor structure. These are conditional results, not an unconditional proof of nature.

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1 Assumptions and claim boundaries

The retained renormalizable baseline is P54PQ-v2, with the declared diagonal center identification in $\text{Spin}(10) \times U(1)_{\text{PQ}}$, and fields

$$3\,16_F + \Phi_{54,\mathbb{R}} + \Sigma_{126,\mathbb{C}} + \phi_{10,\mathbb{C}} + S_{\mathbb{C}}.$$

The name Σ follows the action-card tensor convention; replacing it by its conjugate without changing the Yukawa formula is not allowed. The PQ charges and Yukawa contractions are

$$(q_\Psi, q_\Phi, q_\Sigma, q_\phi, q_S) = (-1, 0, +2, -2, -4), \quad (1)$$

$$-\mathcal{L}_Y = \frac{1}{2}h_{ij}(\Psi_i\Psi_j)_{10}\phi^* + \frac{1}{2}f_{ij}(\Psi_i\Psi_j)_{\overline{126}}\Sigma + \text{h.c.}, \quad h^T = h, \quad f^T = f. \quad (2)$$

The charged-lepton Σ Clebsch is -3 relative to down quarks. A complete numerical spinor Clebsch tensor has not been independently exported: an explicit contraction regression remains useful. Here we use the canonical Yukawa normalization of the action card.

The stored threshold benchmark is

$$\sigma/\omega = 0.1265, \quad v_s/\omega = 0.25, \quad \langle S \rangle = v_s/\sqrt{2}, \quad (3)$$

$$M_U = 8.5347508424 \, 10^{14} \, \text{GeV}, \quad g_U = 0.562602889985, \quad \Delta_D/\omega^2 = 0.0305570302561. \quad (4)$$

P2 determines the hard bosonic curvature on the tree light four-plane,

$$P\Pi_BP/\omega^2 = -0.0202366510950P,$$

but not $Q\Pi_BP$, $Q\Pi_BQ$, or the tadpole-induced vacuum shift. Its light-mass retuning is a projected first-order result, not a complete corrected eigenpair.

Layer P tests this four-dimensional action. Layer U seeks a representation-valued family index. Layer S studies an independently specified soliton action. S does not block P or U. On $\mathbb{CP}^1$, choosing $K^{1/2} = \mathcal{O}(-1)$ and $L_F = \mathcal{O}(3)$ gives

$$\text{ind } D_{L_F} = \int c_1(L_F) = 3, \quad h^0(\mathcal{O}(2)) = 3, \quad h^1(\mathcal{O}(2)) = 0.$$

This derives the count after choosing flux three, not the flux, the 16 representation, anomaly cancellation, or absence of other light states. Likewise the older Route-E upper bound and extra Killing-contact axiom are not an unconditional three-family theorem.

2 Corrected geometry and the exact flavor inequality

2.1 Conjugation before fitting

The reconstructed *tree* light vector has geometric magnitudes

$$(c_1, c_2, c_3, c_4) = (0.6710153956, 0.7414417126, 0.00005897540343, 0.001588153768). \quad (5)$$

Their order is $(\phi_{\text{hol}}, \phi_{\text{anti}}, \Sigma_{\text{hol}}, \Sigma_{\text{anti}})$. They are not pre-labelled up/down Yukawa coefficients. The implementation uses the real generator associated with $-iY$; eigenvalue $-i/2$ selects physical hypercharge $+1/2$. Because h couples ϕ^* and f couples Σ , in

$$Y_d = ah + df, \quad Y_e = ah - 3df, \quad Y_u = bh + ef \quad (6)$$

the declared magnitudes are

$$(|a|, |b|, |d|, |e|) = (c_1, c_2, c_4, c_3). \quad (7)$$

Phases remain arbitrary in the algebra. Positive-real coefficients below are explicitly conditional examples, not a scalar-phase match.

Define $H = ah$, $F = df$, $r = b/a$, $s = ae/(db)$. Eliminating H, F from $Y_d = H + F$, $Y_e = H - 3F$, $Y_u = r(H + sF)$ proves

$$\boxed{Y_u = \frac{r}{4}[(3 + s)Y_d + (1 - s)Y_e].} \quad (8)$$

This identity holds in a common unified flavor basis. Low-energy matrices cannot be independently basis-rotated inside it.

2.2 Sharp phase-uniform norm bound

Write $\rho = |s|$, $B = y_b = \|Y_d\|_2$, $E = y_\tau = \|Y_e\|_2$. The triangle inequality gives

$$y_t \leq \frac{|r|}{4}(|3 + s|B + |1 - s|E).$$

The two factors share one phase. With $x = \cos \arg s$ define

$$F(x) = B\sqrt{9 + \rho^2 + 6\rho x} + E\sqrt{1 + \rho^2 - 2\rho x}.$$

It is concave. Differentiation gives the maximizer, for $\rho > 0$,

$$x_* = \text{clip}_{[-1,1]} \frac{9B^2(1 + \rho^2) - E^2(9 + \rho^2)}{6\rho(E^2 + 3B^2)}. \quad (9)$$

Endpoint inspection covers a stationary point outside the interval. Thus

$$\boxed{y_t \leq U := |r|F(x_*)/4.} \quad (10)$$

For $\rho = 0$, $U = |r|(3B + E)/4$. The bound is sharp given only B, E : proportional rank-one matrices can align the two summands. It is necessary, not sufficient, for a fit.

Historical common-scale inputs transported with a one-loop *SM-only diagnostic* to M_U give

$$(y_t, y_b, y_\tau) = (0.464725012, 0.00634229, 0.00986634).$$

The declared map gives

$$(|r|, \rho) = (1.104954845, 0.033607317), \quad U = 0.00801414124,$$

a factor 57.99 below the target. A global charge-conjugate assignment, used only as a convention stress test, gives $(|r|, \rho) = (0.905014358, 29.7554251)$ and $U = 0.111193537$. It is not an extra fit parameter of the declared action.

The older factor 69.49, repair target $|r| \geq 62.9$, and empty η_Y profile-domain claim are withdrawn. The corrected tree diagnostic still has substantial tension. However, a chosen 10- or 30-percent error box is not a proven bound on missing Pati–Salam running, thresholds, or light-vector rotations. Gaussian likelihood tails also do not vanish: a large lower bound on χ^2 does not empty its mathematical parameter domain. The profile

$$\chi_{\text{prof}}^2(\eta) = \min_{\theta: \eta_Y(\theta) = \eta} \chi^2(\theta)$$

has not been computed.

3 The full light eigenpair: Schur complement and tadpoles

In the tree light/heavy copy decomposition let

$$\mathcal{M}_D^2 = \begin{pmatrix} A & b^\dagger \\ b & C \end{pmatrix}, \quad C > 0.$$

The electroweak degeneracy is implicit. Solving the heavy equation at zero eigenvalue gives $z = -C^{-1}bu$, hence

$$\boxed{A = b^\dagger C^{-1}b, \quad c_{\text{light}} = \frac{c_0 - QC^{-1}b}{\sqrt{1 + \|C^{-1}b\|_2^2}}.} \quad (11)$$

Completing the square gives

$$(u, z)^\dagger \mathcal{M}_D^2(u, z) = (z + C^{-1}bu)^\dagger C(z + C^{-1}bu).$$

The condition is necessary and sufficient for one zero copy with a positive heavy block. When $b = O(\hbar)$, the missed mass is $O(\hbar^2)$ but the vector rotates at $O(\hbar)$.

For example, in ω^2 units,

$$\begin{pmatrix} 0 & 0.001 \\ 0.001 & 0.0305570303 \end{pmatrix}$$

has lowest eigenvalue $-3.26907 \cdot 10^{-5}$. Retuning its upper-left entry to $3.27257 \cdot 10^{-5}$ restores a zero and rotates its vector by 0.0327140 radians. This is an algebraic counterexample to a projected-only criterion, not a computed P54 loop matrix. If E is the full correction and Δ the tree heavy gap, then

$$\tan \theta \leq \frac{\|QE c_0\|_2}{\Delta - \|QEQ\|_2}, \quad \|QEQ\|_2 < \Delta. \quad (12)$$

A Q norm test alone does not supply the numerator.

At fixed renormalized parameters on a physical radial slice,

$$\delta v = -H_{\text{rad}}^{-1} \nabla V_1(v_0), \quad \delta \mathcal{M}_D^2 = \Pi_D + \sum_a \delta v_a \partial_{v_a} \mathcal{M}_{D,0}^2 + \delta \mathcal{M}_{\text{ct}}^2. \quad (13)$$

Alternatively declare fixed-VEV tadpole counterterms. Landau gauge and $\overline{\text{MS}}$ alone do not select this choice. The Schur formula is exact for the assembled truncated matrix; it does not replace omitted two-loop diagrams.

4 Heavy-relaxed quartic: a missing stability check now performed

Work modulo exact gauge and PQ zero modes. Let q be a unit real light vector, z physical heavy coordinates, $C = V^{(2)}|_{\text{heavy}}$, and $J_i = V^{(3)}[e_i, q, q]$. Electroweak symmetry forbids a pure light cubic. Taylor expansion gives

$$V(v_0 + tq + z) = V_0 + \frac{1}{2}z^T C z + \frac{1}{2}t^2 J^T z + \frac{1}{24}V^{(4)}[q^4]t^4 + \dots$$

Since $C > 0$, heavy stationarity gives $z_*(t) = -t^2 C^{-1}J/2 + O(t^3)$. Using $H^\dagger H = t^2/2$,

$$\boxed{\lambda_{\text{EFT}} = \frac{1}{6}V^{(4)}[q^4] - \frac{1}{2}J^T C^{-1}J.} \quad (14)$$

The heavy exchange is non-positive. A full-Hessian pseudoinverse is valid only when J is orthogonal to every zero mode; this is checked.

For the actual 328-real-coordinate potential, all 290 positive scalar modes give

$$\lambda_{\text{direct}} = 0.429206300, \quad \frac{1}{2}J^T C^{-1}J = 0.0200527602, \quad \boxed{\lambda_{\text{EFT}} = 0.409153539 > 0.}$$

Two light directions and derivative steps 0.02, 0.01 agree. The direct paths $v_0 \pm tq - t^2 C^{-1}J/2$ independently give

t	direct-path quartic estimate
0.04	0.409155931
0.02	0.409154138
0.01	0.409153600

The zero-space source norm is below $1.7 \cdot 10^{-16}$. No coupling was fitted to these results. The potential is quartic, so these centered differences extract its polynomial jets up to roundoff. For one complex doublet the quartic invariant is uniquely $(H^\dagger H)^2$. Together with the heavy positive Hessian and no other physical zero modes, this supports local tree stability modulo symmetries. It is numerical evidence, not an interval-arithmetic proof, global boundedness theorem, loop vacuum result, or physical Higgs mass. The quartic itself still needs running and threshold matching.

5 Exact complex hard-Majorana CW projection

Let $z_a = (q_{Ra} + iq_{Ia})/\sqrt{2}$,

$$D(z) = \sum_a z_a Y_a, \quad \mathbb{M}(z) = \begin{pmatrix} 0 & D(z) \\ D(z)^T & M_R \end{pmatrix}, \quad M_R^T = M_R, \quad \det M_R \neq 0.$$

Keep M_R independent of these doublet coordinates. The hard Majorana potential is

$$V_F^{\text{hard}} = -\frac{1}{32\pi^2} \sum_{\text{heavy}} \lambda_i(z)^2 \left[\log \frac{\lambda_i(z)}{\mu^2} - \frac{3}{2} \right].$$

In a Takagi basis, the direct heavy-block correction and mixing through the light block each contribute $(D^\dagger D)_{ii}$, so $\delta\lambda_i = 2(D^\dagger D)_{ii}$. Differentiating $\lambda^2(\log(\lambda/\mu^2) - 3/2)$ gives $2\lambda(\log(\lambda/\mu^2) - 1)$, proving

$$V_F^{(2)} = z^\dagger \Pi z, \quad \boxed{\Pi_{ab} = -\frac{1}{8\pi^2} \text{Tr}(Y_a^\dagger Y_b W)}, \quad (15)$$

$$W = (M_R^\dagger M_R) \left[\log \frac{M_R^\dagger M_R}{\mu^2} - 1 \right].$$

Positive degeneracies follow by continuity. Light seesaw masses are $O(z^2)$ and cannot affect this Hessian. Its real form is

$$\Pi_{\mathbb{R}} = \begin{pmatrix} \Re \Pi & -\Im \Pi \\ \Im \Pi & \Re \Pi \end{pmatrix};$$

discarding imaginary off-diagonal copy mixing is incorrect.

As a conditional scale diagnostic use the published H', F' of Mummidi–Patel [1], $h = H'/a$, $f = F'/d$, $M_R = \sigma f$, $(Y_1, Y_2) = (h, -3f)$, $\mu = g_U \omega$. The up-coupled copies are $(\phi_{\text{anti}}, \Sigma_{\text{hol}})$. With positive-real overlaps,

$$\frac{\Pi_{\text{up}}}{\omega^2} = \begin{pmatrix} 2.2953812 \cdot 10^{-8} & (3.0455254 - 3.2869302i) \cdot 10^{-6} \\ (3.0455254 + 3.2869302i) \cdot 10^{-6} & 9.6808572 \cdot 10^{-4} \end{pmatrix}.$$

Embedding all four complex copies gives

$$\eta_Y = 1.28882418 \cdot 10^{-8}, \quad \delta\xi_F = \eta_Y/w_{10}, \quad (16)$$

$$\|Q(\Pi_F - \delta\xi_F P_{10})Q\|_2 / \omega^2 = 9.68094794 \cdot 10^{-4}, \quad \frac{\|Q(\Pi_F - \delta\xi_F P_{10})Q\|_2}{\Delta_D} = 0.03168157. \quad (17)$$

This is the heavy-neutrino contribution for one externally supplied matrix pair, not a fitted prediction or a total boson–fermion certificate. The actual scalar relative phases are unmatched. Independent finite differences for random complex symmetric matrices on all four real coordinates give relative error $3.7 \cdot 10^{-8}$. Majorana-field variation or new heavy fermions require additional terms.

A Higgs pole instead solves a matched inverse-propagator equation, schematically

$$\det[p^2 Z(\mu) - M^2(\mu) - \Pi(p^2, \mu)] = 0,$$

with consistent self-energy conventions, tadpoles, field matching, and gauge/Nielsen consistency. CW curvature at zero momentum is not this pole.

6 An antisymmetric-only obstruction at fixed geometry

A 120 Yukawa adds antisymmetric tensors: $Y_x = S_x + A_x$, $A_x^T = -A_x$. With the original symmetric sector and fixed r, s ,

$$S_u = \frac{r}{4}[(3+s)S_d + (1-s)S_e].$$

For complex matrices $\|\text{Sym } Y\|_2 \leq (\|Y\|_2 + \|Y^T\|_2)/2 = \|Y\|_2$. Hence $\|S_u\|_2 \leq U$ remains valid.

Lemma 1. *Every complex 3×3 antisymmetric matrix has singular values $(g, g, 0)$.*

Proof. Write $A_{ij} = \epsilon_{ijk} z_k$. Contraction gives $A^\dagger A = |z|^2 I - z z^\dagger$, with eigenvalues $|z|^2, |z|^2, 0$. $\square$

Proposition 1. *A three-family, fixed-geometry, antisymmetric-only repair obeys*

$$\boxed{y_t - y_c \leq 2 \|S_u\|_2 \leq 2U.} \quad (18)$$

Proof. The singular-value perturbation inequality gives $|\sigma_j(A_u + S_u) - \sigma_j(A_u)| \leq \|S_u\|_2$. Subtract the first two singular values, equal for A_u . A sum of antisymmetric tensors is still antisymmetric. $\square$

The target requires $\|S_u\|_2 \geq 0.231588278$, versus $U = 0.008014141$ (or 0.111193537 in the conjugate stress test). Two hundred random complex tests reproduce the pairing and bound. This excludes only the stated fixed-geometry mechanism, not a full 120_H extension whose potential changes the light vector. Naming a composite 120 therefore does not yet solve the problem.

7 Constructive route I: inverse scalar–flavor feasibility

Suppose an effective fit supplies H, F, r, s . Impose declared operator norm caps $\|h\|_2 \leq h_*$, $\|f\|_2 \leq f_*$. With no additional light components,

$$(1 + |r|^2)|a|^2 + (1 + |rs|^2)|d|^2 = 1.$$

The caps imply $|a|^2 \geq \|H\|_2^2 / h_*^2$, $|d|^2 \geq \|F\|_2^2 / f_*^2$. Conversely any leftover nonnegative normalization can be distributed to these weights. Thus

$$\boxed{(1 + |r|^2) \frac{\|H\|_2^2}{h_*^2} + (1 + |rs|^2) \frac{\|F\|_2^2}{f_*^2} \leq 1} \quad (19)$$

is necessary and sufficient for normalized *magnitudes* under only these caps. Nonzero denominators are understood; zero tensors are treated by continuous limits. This does not assert scalar realization.

A weaker necessary gate follows from $Y_u = rY_d + (e - rd)f$. With $w_{126} = |d|^2 + |e|^2$, $|e - rd| \leq \sqrt{1 + |r|^2} \sqrt{w_{126}}$, so, when $y_t > |r|y_b$,

$$w_{126} \geq \frac{(y_t - |r|y_b)^2}{f_*^2(1 + |r|^2)}. \quad (20)$$

At the permissive $f_* = 4\pi$ and declared diagnostic r , $w_{126} \geq 5.97366 \cdot 10^{-4}$ versus tree $2.52571 \cdot 10^{-6}$. The corresponding minimum amplitude rotation is

$$\arcsin \sqrt{w_{\text{required}}} - \arcsin \sqrt{w_{\text{tree}}} = 0.0228543 \text{ rad.}$$

Thus the tiny 126 weight, not small decimal residuals, is structural. This rotation is not excluded by an uncomputed off-diagonal loop block; its sign and size are not predicted by the lower bound. The cap 4π is not a perturbative-convergence certificate.

At fixed VEVs, the action Hessian is affine in real invariant couplings:

$$\mathcal{M}_D^2(p) = M_0 + \sum_{\alpha} p_{\alpha} M_{\alpha}.$$

For a fixed candidate normalized complex c , set $Q_c = I - cc^\dagger$ and impose

$$\mathcal{M}_D^2(p)c = 0, \quad Q_c \mathcal{M}_D^2(p) Q_c \succeq \Delta Q_c, \quad \nabla V(v; p) = 0, \quad (21)$$

plus positivity of other physical irreps. These are linear equalities and semidefinite inequalities in p . The outer search over c remains nonconvex; variable VEVs and loops are also not affine. Every feasible inner solution must pass the relaxed quartic, global/competing-vacuum tests, perturbativity, full loop eigenpair and P2 threshold replay. This is a targeted inverse problem rather than a new lattice scan.

8 Constructive route II: PQ-preserving mirror-10

8.1 The symmetry allows the needed effective operator

The renormalizable truncation forbids $(\Psi\Psi)_{10}\phi$, but permits

$$-\Delta\mathcal{L} = \frac{S^*}{2\Lambda} C_{ij} (\Psi_i \Psi_j)_{10}\phi + \text{h.c.}, \quad C^T = C. \quad (22)$$

Its PQ charge is $+4 - 1 - 1 - 2 = 0$. Set $g = C\langle S^* \rangle/\Lambda$. The conjugate overlaps give

$$Y_u = bh + ef + a^*g, \quad (23)$$

$$Y_d = ah + df + b^*g, \quad Y_e = ah - 3df + b^*g. \quad (24)$$

Now $H = ah + b^*g$, $F = df$. Elimination proves

$$\boxed{Y_u - \frac{r}{4}[(3+s)Y_d + (1-s)Y_e] = \kappa_{\text{mir}}g, \quad \kappa_{\text{mir}} = \frac{|a|^2 - |b|^2}{a}.} \quad (25)$$

This new symmetric direction escapes the antisymmetric obstruction. It fails when $|a| = |b|$. At the positive-real declared point, $\kappa_{\text{mir}} = -0.148244217$. A necessary size is $\|g\|_2 \geq (y_t - U)/|\kappa_{\text{mir}}|$.

An explicit third-family toy uses matrices proportional to $P_3 = \text{diag}(0, 0, 1)$ and

$$H_3 = (3y_b + y_\tau)/4, \quad F_3 = (y_b - y_\tau)/4, \quad (26)$$

$$f_3 = F_3/d, \quad g_3 = (y_t - rH_3 - ef_3)/\kappa_{\text{mir}}, \quad h_3 = (H_3 - bg_3)/a. \quad (27)$$

It gives

$$(h_3, f_3, g_3) = (3.41539797, -0.55474038, -3.08124196)$$

and reproduces the three targets with residual below $5 \cdot 10^{-16}$. All lighter families, mixing and neutrino observables are absent: this is an algebraic escape, not a viable fit. The down/lepton cancellations also require a sensitivity audit.

8.2 Mediator construction with exact VEV mixing

Add left-handed Weyl fields $X \sim 16$ with $q_X = +3$ and $\bar{X} \sim \bar{16}$ with $q_{\bar{X}} = -3$, with

$$-\mathcal{L}_{\text{med}} = M\bar{X}X + \kappa_i S^* \bar{X} \Psi_i + y_i (\Psi_i X)_{10}\phi + \text{h.c.} \quad (28)$$

Every vertex is neutral. Leading-order elimination gives

$$g_{ij} = -\frac{\langle S^* \rangle}{M} (y_i \kappa_j + y_j \kappa_i).$$

One pair gives rank at most two; aligned vectors give rank one. It is not a completion of arbitrary full-rank C with one pair. The same elimination yields

$$Z_\Psi = I + \frac{|\langle S \rangle|^2}{|M|^2} \kappa^* \kappa^T + \dots,$$

which cannot be dropped for non-small mixing.

In the aligned one-family case rephase the VEV mixing real and set $\tan \theta = |\kappa \langle S \rangle|/M$. The light state has $\Psi = \cos \theta \chi + \dots$ and $X = -\sin \theta \chi + \dots$, giving exactly

$$h = h_{UV} \cos^2 \theta, \quad f = f_{UV} \cos^2 \theta, \quad g = -y_{UV} \sin 2\theta, \quad M_{\text{phys}} = M / \cos \theta. \quad (29)$$

The phase of y_{UV} supplies the sign of g . Balancing the largest toy vertex norms gives

$$\tan \theta = \frac{|g_3|}{2|h_3|}, \quad \max(|h_{UV}|, |y_{UV}|) = |h_3| + \frac{|g_3|^2}{4|h_3|}.$$

Here $\theta = 0.42375249$ and $|h_{UV}| = |y_{UV}| = 4.11034$; f must also be divided by $\cos^2 \theta$. Choosing illustrative bare $M = g_U \omega$ gives $|\kappa| = 1.43559$ and $M_{\text{phys}} = 0.617192 \omega$. The vertex loop parameter is about 0.107 even before representation multiplicities: weak perturbativity is not established. New thresholds, beta functions, CW terms, scalar counterterms and Landau-pole checks are obligatory. Exact mixing avoids falsely using an uncontrolled dimension-five VEV expansion.

The pair adds zero PQ-gauge anomaly, $3T(16) - 3T(\overline{16}) = 0$; gravitational and cubic PQ anomalies also cancel. It descends under the existing diagonal center quotient. The baseline physical $N_{\text{DW}} = 3$ is unchanged, so axion cosmology is not repaired. The separate charge-+2 10_F branch must not be silently combined with this proposal. Rank-one flavor from vectorlike 16s has precedent [4]; the contribution here is a specific PQ-compatible opposite-conjugation mechanism and its obstruction tests, not a claim of priority for the general idea.

9 Einstein–Cartan torsion and central twists

Write a metric-compatible connection as $\omega = \omega(e) + K$. The Einstein–Hilbert action is quadratic in algebraic contorsion up to a divergence; minimal fermions source its axial component. In abstract notation,

$$\mathcal{L}_s = \frac{1}{2} s A s + s J \implies s = -A^{-1} J, \quad \Delta \mathcal{L} = -\frac{1}{2} J A^{-1} J.$$

In conventional Einstein–Cartan normalization the induced axial current-current coefficient has magnitude $3/(16M_P^2)$ [5]; its displayed sign follows the chosen curvature, metric and axial conventions. It is a dimension-six interaction, not an independent renormalizable symmetric Yukawa matrix, and preserves local gauge/PQ selection rules. Loop renormalization of allowed operators depends on spurions and matching; NDA does not give a universal nonzero δY . For $M_P = 2.435 \cdot 10^{18}$ GeV and the stored M_U ,

$$M_U^2/M_P^2 = 1.2285 \cdot 10^{-7}, \quad (3/16)M_U^2/M_P^2 = 2.3035 \cdot 10^{-8}.$$

An extra loop factor gives an NDA size $1.46 \cdot 10^{-10}$, only a suppression diagnostic. Low-scale propagating torsion or a condensate is a new theory with kinetic-sign, unitarity, symmetry-breaking, gap and spectrum requirements. Inferring a torsion mass below M_U from a contact interaction, then applying that contact EFT at M_U , is inconsistent.

A central twist is distinct from spacetime contorsion and from torsion cohomology. The center of $\text{Spin}(10)$ is discrete; its Lie algebra has no extra central generator. A global quotient can change bundles and line operators without adding local Yukawa tensors. The Nieh–Yan identity is

$$d(e^a \wedge T_a) = T^a \wedge T_a - e^a \wedge e^b \wedge R_{ab}.$$

It does not supply an arbitrary quantized QCD anomaly shift. The 2024 boundary/renormalization analysis [6] emphasizes counterterms and renormalization conditions for torsional anomalies. A PQ repair must compute the actual fermion measure and vacuum identifications. Since the gauge-singlet S has charge -4 , a gauge center cannot compensate its phase except when $e^{-4i\alpha} = 1$. The existing quotient already accounts for that periodicity; minimal EC does not turn the stored $N_{\text{DW}} = 3$ into one.

10 Recent literature and execution order

The 2023 $54 + 126 + 10$ model of Haba–Shimizu–Yamada [2] keeps both conjugate 10 Yukawas by not imposing visible-sector PQ charges. It is a close comparison for a three-symmetric-tensor branch, not evidence of a fit for our frozen two-tensor PQ branch. The mirror operator restores that structural freedom after PQ breaking with explicit mediator and matching costs. The 2025 spontaneous-CP study [3], in a different $45 + 126 + 10$ model, illustrates an extra-light-doublet option with correlated flavor constraints. Its conclusion cannot be transplanted to P54PQ without a separate action and flavor analysis.

The next order is:

1. Retain the corrected conjugation map and positive relaxed quartic. Declare a tadpole scheme, compute all complex doublet P/Q blocks, and obtain the corrected light eigenvector, not only its projected mass.
2. Couple Pati–Salam Yukawa running and finite matching to the inverse scalar–flavor problem (19)–(21). Profile η_Y from actual matrices; one scalar nuisance cannot replace a matrix-valued spectrum constraint.
3. If the baseline has no viable region, compare the rank-restricted mirror-10 completion, two-doublet EFT, and enlarged scalar sector. Recalculate parameter costs, perturbativity, thresholds and cosmology.
4. Promote neutrino intervals, proton amplitudes and Higgs poles only after their fit/matching gates. Keep family UV and soliton mathematics independent of these tasks.

No calculation here proves nature selects this gauge group, representations, PQ charges, couplings or vacuum. The defensible first-principles program derives consequences from explicit assumptions and tests the resulting models.

A Reproduction and status ledger

The authoritative ledger is `route_f/output/p54_theory_audit.json`; the compatibility P3 output now uses the same corrected schema. The superseded conjugation/real-slice implementation cannot regenerate its earlier exclusion. From the repository root:

```
python3 route_f/code/extract_p54_p3_light_overlaps.py
python3 route_f/code/verify_p54_relaxed_light_quartic.py
python3 route_f/code/verify_p54_theory_audit.py
```

The first two require JAX double precision, NumPy and SciPy; this run used Python 3.9, JAX/JAXlib 0.4.30, NumPy 2.0.2, SciPy 1.13.1. The third needs only NumPy/SciPy. Source hashes are stored. Overlap extraction passes 5/5, relaxed quartic 7/7, and independent audit 14/14 checks. These validate the stated implementations, not the entire theory. Still uncompleted: full loop eigenpair, global flavor/seesaw fit, η_Y profile, fitted mirror UV branch, full Q certificate, and physical Higgs pole.

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